

Math 208 Conceptual Problems #5 (Sections 5.1-5.2) - **SOLUTIONS**

(1) Find the determinant of the matrix

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ -2 & 0 & 4 \end{bmatrix}^3 \begin{bmatrix} 8 & 0 & 3 \\ -1 & 1 & 1 \\ 0 & 2 & 4 \end{bmatrix}^{-1}$$

Solution. There are lots of ways to do this. Here's one. First, let's compute the determinant of the first matrix:

$$\begin{aligned} \det \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ -2 & 0 & 4 \end{bmatrix} &= 2 \cdot \det \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \quad (\text{factoring 2 from the 3rd row}) \\ &= 2 \cdot \det \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 3 & 4 \end{bmatrix} \quad (\text{row operation } R_1 + R_3 \rightarrow R_3, \text{ det unchanged}) \\ &= 2 \cdot 1 \cdot \det \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} \quad (\text{cofactor expansion down first column}) \\ &= 2 \cdot (4 - 3) = 2. \end{aligned}$$

Next,

$$\begin{aligned} \det \begin{bmatrix} 8 & 0 & 3 \\ -1 & 1 & 1 \\ 0 & 2 & 4 \end{bmatrix} &= 8 \det \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} \quad (\text{cofactor expansion along first row}) \\ &= 8(4 - 2) + 3(-2 - 0) = 10. \end{aligned}$$

Now using the rules $\det(AB) = \det(A)\det(B)$ and $\det(A^{-1}) = (\det(A))^{-1}$, the final determinant is $2^3 \cdot 10^{-1} = \frac{8}{10}$.

(2) Compute $\det(5A^3)$ for the matrix A below.

$$A = \begin{bmatrix} 0 & 2 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 3 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 5 & 6 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 6 & 5 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Solution. Since $5A^3 = (5I)(A^3)$ we have that $\det(5A^3) = \det(5I)\det(A^3) = 5^{10}(\det(A))^3$. So the work involved is to compute $\det(A)$.

The full computation is messy to show, but easy. Do cofactor expansion by sequentially picking a row or column that has only one non-zero entry in it. For instance if you expand along the first column we see the determinant has to be

$$-1 \cdot \det \begin{bmatrix} 2 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 3 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 4 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 5 & 6 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 6 & 5 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 4 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = (-1)(2) \det \begin{bmatrix} 3 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 4 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 5 & 6 & 1 & 0 & 1 & 0 \\ 0 & 0 & 6 & 5 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 4 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

and so on. Check that $\det(A) = 11 \times (24)^2$. Therefore, $\det(5A^3) = 5^{10}11^3(24)^6$.

- (3) This exercise uses the notion of *cofactors* of a matrix defined in Section 5.1. Recall that if A is an $n \times n$ matrix, then M_{ij} denotes the matrix obtained from A by deleting the i th row and j th column of A .

$$\text{Let } A = \begin{bmatrix} -2 & 0 & 2 \\ 1 & 1 & 1 \\ 3 & -1 & 5 \end{bmatrix}.$$

- (a) Compute all nine cofactors of A , as well as $\det(A)$. Let B be the 3×3 matrix containing the cofactors, with each entry multiplied by the appropriate $\pm$ sign. So the ij -entry of B is $(-1)^{i+j} \det(M_{ij})$.

Solution. The first cofactor is $(-1)^{1+1} \det \begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix} = 6$, from deleting the first row and column. Computing all cofactors we get

$$B = \begin{bmatrix} 6 & -2 & -4 \\ -2 & -16 & -2 \\ -2 & 4 & -2 \end{bmatrix}$$

To compute $\det(A)$, I used a column operation, $\text{col}_1 + \text{col}_3 \rightarrow \text{col}_3$:

$$\det \begin{bmatrix} -2 & 0 & 2 \\ 1 & 1 & 1 \\ 3 & -1 & 5 \end{bmatrix} = \det \begin{bmatrix} -2 & 0 & 0 \\ 1 & 1 & 2 \\ 3 & -1 & 8 \end{bmatrix} = -2 \det \begin{bmatrix} 1 & 2 \\ -1 & 8 \end{bmatrix}$$

(by cofactor-expanding along the top row). This works out to $\det(A) = -20$.

- (b) Compute AB^T . You should get a diagonal matrix with the same number in every diagonal entry, in other words, a multiple of the identity matrix. What multiple is it (in terms of A)?

Solution. Computing AB^T gives

$$\begin{bmatrix} -2 & 0 & 2 \\ 1 & 1 & 1 \\ 3 & -1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 6 & -2 & -2 \\ -2 & -16 & 4 \\ -4 & -2 & -2 \end{bmatrix} = \begin{bmatrix} -20 & 0 & 0 \\ 0 & -20 & 0 \\ 0 & 0 & -20 \end{bmatrix}.$$

So, in terms of the identity matrix, this is $-20 \cdot I$. This appears to be $\det(A) \cdot I$.

(c) Fill in the blank (with a scalar) to make this equation true:

$$AB^T = (\text{ ? }) \cdot I, \text{ therefore } A^{-1} = \frac{1}{(\text{ ? })} \cdot B^T.$$

Solution. The scalar is $\det(A)$, so $AB^T = \det(A)I$, or $\frac{1}{\det(A)}AB^T = I$. (Bear in mind that $\det(A)$ is a number, not a matrix.) If we multiply on the left by A^{-1} , we get $\frac{1}{\det(A)}B^T = A^{-1}$.

(d) Compute the matrix $B^\top$ when A is the 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

and then use it to compute a formula for A^{-1} . Does this agree with the formula for A^{-1} given in the textbook?

Solution. In this case the matrix $B = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$ and hence $B^\top = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. The determinant of A is $ad - bc$ by cofactor expansion. Therefore, by part (c), we get that

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

which agrees with the formula for the inverse of a 2×2 matrix given in the book.

Comment. The formula $A^{-1} = \frac{1}{\det(A)}B^T$ is bad for computational purposes because it requires computing n^2 separate determinants (the entries of B), each of size $(n-1) \times (n-1)$. And computing $\det(A)$ as well. Computing that many determinants by hand is error-prone and slow, compared to just row reducing. **What about with computers?** Computing a single determinant really efficiently takes roughly n^3 steps, so this method takes $(n-1)^2 \cdot n^3 \approx n^5$ steps. For a matrix with a million (10^6) rows and columns, this means 10^{30} steps. By contrast, computing A^{-1} directly by row reduction takes roughly n^3 steps, total. So, for a matrix with one million rows (10^6), around 10^{18} steps, which is doable (in 2017, a modern supercomputer can do around 10^{15} arithmetic operations per second, so it would take 1000 seconds to do 10^{18} operations – about 15 minutes. For 10^{30} operations, it would take 10^{12} seconds, which is around 30000 years.)

(4) Consider the plane P in $\mathbb{R}^3$ spanned by the vectors $(1, 0, 1)$ and $(2, -1, 0)$.

(a) Express P as the solution set of a linear system in the variables x, y, z .

Solution. Since the two vectors are linearly independent, they will span a 2-dimensional plane in $\mathbb{R}^3$. Therefore, we expect this plane to be the solution set of a single linear equation in x, y, z of the form $ax + by + cz = 0$. To find the equation of the plane we need to find a, b, c . Since the plane contains the two given vectors, plugging them in we should have

$$a + c = 0 \quad \text{and} \quad 2a - b = 0.$$

Solving we get that $c = -a$ and $b = 2a$. Therefore, $(a, b, c) = a(1, 2, -1)$. In particular, choosing $a = 1$, we see that the equation of the plane P is $x + 2y - z = 0$.

(b) Compute the following determinant:

$$\det \begin{bmatrix} x & 1 & 2 \\ y & 0 & -1 \\ z & 1 & 0 \end{bmatrix}.$$

Solution. Using cofactor expansion along the first column, we get the above determinant to be $x(1) - y(-2) + z(-1) = x + 2y - z$.

(c) Check that the equation obtained by setting the determinant you just computed to zero has the same solutions as the linear system from (1). Why does this happen?

Solution. The determinant in (b) is 0 exactly when the columns of the matrix in (b) are linearly dependent. In other words, the set of all (x, y, z) for which the determinant is 0 is exactly the set of vectors that lie in the span of the second and third columns of the matrix, which is exactly the set of all (x, y, z) in $\mathbb{R}^3$ that lie on the plane P . This is why setting the determinant of the matrix from (b) to 0 gives the equation of the plane P .

(d) Can you now explain the formula for computing the cross product of two vectors in $\mathbb{R}^3$?

Solution. Look up your calculus book to check that the cross product of the vectors (a, b, c) and (d, e, f) is $(bf - ce, cd - af, ae - bd)$. Notice that these entries are exactly the cofactors of x, y, z in the matrix

$$\det \begin{bmatrix} x & a & d \\ y & b & e \\ z & c & f \end{bmatrix}.$$

Therefore, by the previous part, we see that the equation of the plane spanned by (a, b, c) and (d, e, f) is

$$(bf - ce)x + (cd - af)y + (ae - bd)z = 0.$$

This means that $(bf - ce, cd - af, ae - bd)$ is the normal vector to the plane spanned by (a, b, c) and (d, e, f) which is (up to sign) what the cross product of (a, b, c) and (d, e, f) is.

(e) Generalize the process described in this example to $\mathbb{R}^4$. Take 3 linearly independent vectors (of your choice) and express the plane they span as the set of solutions to a linear equation by computing a determinant as above.

Solution. Suppose we are given 3 linearly independent vectors in $\mathbb{R}^4$, say $(a, b, c, d), (e, f, g, h), (i, j, k, l)$. Then by the exact same reasoning as in (c) we see that the equation of the plane they span would be

$$\det \begin{bmatrix} x & a & e & i \\ y & b & f & j \\ z & c & g & k \\ w & d & h & l \end{bmatrix} = 0.$$

where the variables are now x, y, z, w .

Note that to compute a determinant we need a square matrix, so this method only works (in $\mathbb{R}^n$) when we have a plane spanned by $n - 1$ vectors. If the vectors are linearly dependent, then they actually span a plane of dimension lower than $n - 1$ and we need more than one equation to express the plane. The determinant method only provides one equation. Therefore the above procedure to compute the equation of a plane in $\mathbb{R}^n$ works only when the dimension of the plane is $n - 1$. In this case we set up the matrix by putting n variables in the first column and using a basis of the plane to fill the remaining columns to get an $n \times n$ matrix whose determinant we can calculate.

- (5) (Determinants and geometry) This exercise assumes that students know about the connection between geometry and volumes of parallelopipeds. This is in Section 5.3 which is not required in the syllabus, but is worthwhile to know.

- (a) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be counter clockwise rotation by $\pi/3$, i.e. $T(\mathbf{x})$ is the vector obtained by rotating $\mathbf{x}$ counter clockwise by $\pi/3$ around $\mathbf{0}$. Without computing any matrices, what would you expect $\det(T)$ to be? (Does T make areas larger or smaller?)

Now check your answer by using the fact that the matrix for counter clockwise rotation by θ is

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}.$$

Solution. Rotation does not change volumes at all, so we expect $\det(T) = \pm 1$. Also, rotation does not involve reflecting (taking a mirror image), so $\det(T)$ should probably be positive. So, we should guess $\det(T) = 1$. Check:

$$\det \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} = \cos^2(\theta) + \sin^2(\theta) = 1.$$

- (b) Same question as (a), only this time let T be the transformation that reflects $\mathbb{R}^2$ over the line $y = x$. That is, $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} y \\ x \end{bmatrix}$.

Guess what $\det(T)$ should be, then check by finding the matrix for T and computing its determinant.

Solution. Again, areas and volumes should not change, so $\det(T)$ should be ± 1 . Since T does involve a reflection, we expect $\det(T) = -1$.

The matrix for T is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (because $T(\mathbf{e}_1) = \mathbf{e}_2$ and $T(\mathbf{e}_2) = \mathbf{e}_1$). So, the determinant is:

$$\det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 0 - 1 = -1.$$

- (c) Rotation matrices in $\mathbb{R}^3$ are more complicated than in $\mathbb{R}^2$ because you have to specify an axis of rotation, which could be any line through the origin. Nonetheless, what would you expect $\det(T)$ to be? Look up the “basic 3D rotation matrices” on Wikipedia (https://en.wikipedia.org/wiki/Rotation_matrix#In_three_dimensions) and compute $\det(A)$ for each one.

Solution. Same reasoning as in part (a): we should guess $\det(T) = 1$. Wikipedia gives these three rotation matrices:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix} \quad \begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \quad \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

You can check that $\det(A) = 1$ in each case. (Use cofactor expansion along the row containing the 1.)

- (d) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be projection onto the xy -plane, so $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$. What is $\det(T)$? Guess first and then check using a matrix.

Solution. The transformation T flattens everything! So we should expect $\det(T) = 0$. Also, we know that T is **not invertible**, because it is not onto (since $\text{range}(T) = \text{the } xy\text{-plane only}$). This forces $\det(T) = 0$. To check (this is just for practice – **we know** $\det(T) = 0$ from the fact that T isn’t invertible!), we find the matrix for T . We have $T(\mathbf{e}_1) = \mathbf{e}_1$ and $T(\mathbf{e}_2) = \mathbf{e}_2$, but $T(\mathbf{e}_3) = \mathbf{0}$. So the matrix is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Since the matrix is diagonal, $\det(T)$ is the product of the diagonal entries, so $\det(T) = 1 \cdot 1 \cdot 0 = 0$.

(6) (Determinants and interpolation)

Suppose we want to make a quadratic polynomial

$$y = f(x) = a_0 + a_1x + a_2x^2$$

that passes through three specified points $\mathbf{p}_1 = (p_1, q_1)$, $\mathbf{p}_2 = (p_2, q_2)$, $\mathbf{p}_3 = (p_3, q_3)$. Consider the equation

$$0 = \det \begin{bmatrix} 1 & x & x^2 & y \\ 1 & p_1 & p_1^2 & q_1 \\ 1 & p_2 & p_2^2 & q_2 \\ 1 & p_3 & p_3^2 & q_3 \end{bmatrix}.$$

The determinant above implicitly gives an equation $y = f(x)$ (it’s easy to solve for y since no y^2, y^3 , etc terms appear).

- (a) Write out the matrix above, using $\mathbf{p}_1 = (0, 0)$, $\mathbf{p}_2 = (1, 1)$, $\mathbf{p}_3 = (3, 5)$ for the constants p_i, q_i , but leaving x and y as variables. Solve the equation $\det(A) = 0$ to get $y = \text{a quadratic polynomial in } x$. Check directly that the parabola passes through $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$.

Solution. The equation is

$$0 = \det \begin{bmatrix} 1 & x & x^2 & y \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 5 \end{bmatrix} = -1 \cdot \det \begin{bmatrix} x & x^2 & y \\ 1 & 1 & 1 \\ 3 & 9 & 5 \end{bmatrix}$$

(from expanding along the second row). Expanding along the first row, we get

$$\begin{aligned} 0 &= (-1) \cdot \left(x \det \begin{bmatrix} 1 & 1 \\ 9 & 5 \end{bmatrix} - x^2 \det \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix} + y \det \begin{bmatrix} 1 & 1 \\ 3 & 9 \end{bmatrix} \right) \\ &= -(-4x - 2x^2 + 6y) \end{aligned}$$

giving $y = \frac{2}{3}x + \frac{1}{3}x^2$.

Now, if we check, the equation is satisfied by $(0, 0)$, $(1, 1)$ and $(3, 5)$. A miracle!

- (b) Why does part (a) succeed? Examine the matrix A from part (a). If you plug in $(x, y) = \mathbf{p}_1 = (0, 0)$ to the first row of A , the first two rows will become the same. So, by the ‘repeated rows’ rule, the equation $\det(A) = 0$ must be true for those specific x, y values. What does this mean about the polynomial $y = f(x)$?

What about if you plug in $(x, y) = (1, 1)$ or $(3, 5)$? Why (in terms of determinants) must the equation $y = f(x)$ be satisfied by these points?

Solution. The logic here is saying to examine the equation $\det(A) = 0$ directly as a determinant, *without* expanding it out:

$$0 = \det \begin{bmatrix} 1 & x & x^2 & y \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 5 \end{bmatrix}$$

This is a valid equation relating x and y . We can test if it’s satisfied by particular (x, y) values by plugging them in. If we plug in $(x, y) = (0, 0)$, we get

$$0 = \det \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 5 \end{bmatrix}$$

which is a *true equation* because the determinant visibly has a repeated row. So $(0, 0)$ satisfies the equation. Similarly, if we plug in $(x, y) = (1, 1)$, we get

$$0 = \det \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 5 \end{bmatrix}$$

which is true because the first and third row are the same. Lastly, if we plug in $(x, y) = (3, 5)$, we get

$$0 = \det \begin{bmatrix} 1 & 3 & 9 & 5 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 5 \end{bmatrix}$$

and this is true because the first and last rows are the same.

- (c) How could you use a determinant to make a cubic polynomial that passes through 4 given points? (It should require a 5×5 determinant.)

Solution. Use the equation

$$0 = \det \begin{bmatrix} 1 & x & x^2 & x^3 & y \\ 1 & p_1 & p_1^2 & p_1^3 & q_1 \\ 1 & p_2 & p_2^2 & p_2^3 & q_2 \\ 1 & p_3 & p_3^2 & p_3^3 & q_3 \\ 1 & p_4 & p_4^2 & p_4^3 & q_4 \end{bmatrix}.$$

This is implicitly an equation of the form $y = a_0 + a_1x + a_2x^2 + a_3x^3$. The right hand side is a cubic polynomial in x . The equation is satisfied if we plug in $(x, y) = (p_i, q_i)$, because then there is a repeated row, guaranteeing that $\det = 0$.

- (7) Let $A(t) = b_1 \cos(\omega t) + b_2 \sin(\omega t)$ be the ambient temperature in your kitchen, which varies sinusoidally as function of time t . Assume $A(t)$ is known — that is, the values of b_1 , b_2 , and ω have been measured.

You have just cooked a cup of noodles. *Newton's Law of Cooling* states that the temperature function $y(t)$ of the noodles satisfies

$$y'(t) = -k(y(t) - A(t))$$

with k a (known) positive constant. The “steady-state” solution (which y approaches in the limit as t increases) is of the form

$$y(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t).$$

Use Newton's Law of Cooling to write c_1 and c_2 in terms of b_1, b_2 . Write the linear relations you found as a system of equations with kb_i on the right side. Then use an inverse matrix to find a formula for the steady-state solution.

Solution. Using calculus, compute that $y'(t) = -c_1\omega \sin(\omega t) + c_2\omega \cos(\omega t)$. On the other hand, we are given that

$y'(t) = -k(y(t) - A(t)) = -k(c_1 - b_1) \cos(\omega t) - k(c_2 - b_2) \sin(\omega t)$. This means that

$$-c_1\omega = -k(c_2 - b_2), \quad c_2\omega = -k(c_1 - b_1)$$

which implies that

$$\begin{aligned} kc_1 + \omega c_2 &= kb_1 \\ -\omega c_1 + kc_2 &= kb_2 \end{aligned}$$

or equivalently,

$$\begin{bmatrix} k & \omega \\ -\omega & k \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} kb_1 \\ kb_2 \end{bmatrix}.$$

Now multiply both sides by the inverse of the matrix on the left to get

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \frac{1}{\omega^2 + k^2} \begin{bmatrix} k & -\omega \\ \omega & k \end{bmatrix} \begin{bmatrix} kb_1 \\ kb_2 \end{bmatrix}$$

Now we can substitute for c_1, c_2 in the formula for $y(t)$ to get

$$y(t) = \frac{k}{\omega^2 + k^2} [(kb_1 - \omega b_2) \cos(\omega t) + (\omega b_1 + kb_2) \sin(\omega t)].$$

(8) Let

$$f(t) = x_1 \sin(t) + x_2 \cos(t) + x_3 t \sin(t) + x_4 t \cos(t)$$

and

$$f'(t) = y_1 \sin(t) + y_2 \cos(t) + y_3 t \sin(t) + y_4 t \cos(t),$$

where the x_i and y_i are coefficients. Find a matrix A expressing the vector $\mathbf{y} = (y_1, y_2, y_3, y_4)$ as $\mathbf{y} = A\mathbf{x}$ where $\mathbf{x} = (x_1, x_2, x_3, x_4)$. What is the interpretation of $A^2\mathbf{x}$ and $A^3\mathbf{x}$?

Solution. We first compute the derivative of $f(t)$ to see that

$$f'(t) = (x_1 + x_4) \cos(t) + (x_3 - x_2) \sin(t) - x_4 t \sin(t) + x_3 t \cos(t).$$

Now comparing this to the $f'(t)$ given to us we see that

$$y_1 = x_3 - x_2, \quad y_2 = x_1 + x_4, \quad y_3 = -x_4, \quad y_4 = x_3.$$

Writing this with matrix notation we get

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \mathbf{A}\mathbf{x}.$$

Since $f'(t)$ has the same form as $f(t)$, we can calculate higher derivatives of $f(t)$ by using the matrix A . For example, if

$f''(t) = z_1 \sin(t) + z_2 \cos(t) + z_3 t \sin(t) + z_4 t \cos(t)$, then $\mathbf{z} = \mathbf{A}\mathbf{y} = \mathbf{A}^2\mathbf{x}$. Check this by actually computing $f''(t)$ and using the method we just used. Similarly, if

$f'''(t) = w_1 \sin(t) + w_2 \cos(t) + w_3 t \sin(t) + w_4 t \cos(t)$, then $\mathbf{w} = \mathbf{A}\mathbf{z} = \mathbf{A}^2\mathbf{y} = \mathbf{A}^3\mathbf{x}$.

(9) Let $A = \begin{bmatrix} -4 & 3 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix}$.

(a) Compute $\det(A)$ and $\det(B)$.

Solution. $\det(A) = -1$ and $\det(B) = 1$.

(b) Using facts about determinants and without computing AB , argue that the columns of AB form a basis of $\mathbb{R}^2$.

Solution. Since $\det(AB) = \det(A)\det(B) = -1$, we know that the columns of AB are linearly independent. Since there are two columns in AB , these columns must be a basis for $\mathbb{R}^2$.

(c) Compute A^{-1} and B^{-1} .

Solution. $A^{-1} = \begin{bmatrix} -1 & 3 \\ -1 & 4 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 3 & -1 \\ 1 & 0 \end{bmatrix}$.

(d) Compute the coordinates of the point $(1, 1)$ in the basis formed by the columns of AB . Express your computation abstractly in terms of A and B before you use the specific A and B given above.

Solution. We are looking for (x, y) such that

$$AB \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

This means that

$$\begin{bmatrix} x \\ y \end{bmatrix} = (AB)^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = B^{-1}A^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 & 5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

(e) What is the linear transformation T that will undo the change of basis in (c) bringing your answer in (c) back to $(1, 1)$? Write T with domain, codomain and matrix. Express your computation abstractly in terms of A and B before you use the specific A and B given above. Check that it works.

Solution. This map must be $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = AB \begin{bmatrix} x \\ y \end{bmatrix}$. Let's check if

this will send $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ back to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

$$AB \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 & 5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$